

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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### COURSE OUTCOME COVERED

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

*Assessment Tool Weightage: 35%*

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**QUESTIONS: 5**

**Duration : 1 Hour**  
**Total Marks:50**

Annexure A

### Analytical Problem Solving

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#### ***Code of Conduct:***

*I \_\_\_\_\_ P.NAGENDRA REDDY (192472399) \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

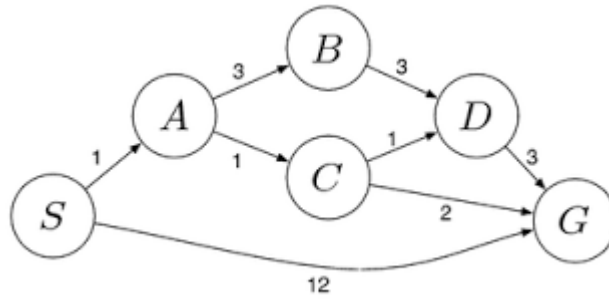
NAGENDRA REDDY

## Annexure A

### **Analytical Problem Solving**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
  - i. What are the percept's for this agent?
  - ii. Characterize the operating environment.
  - iii. What are the actions the agent can take?
  - iv. How can one evaluate the performance of the agent?
  - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



### RUBRICS FOR CALCULATION:

| Criteria                  | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                      | Level 2 (Developing)                       | Level 1 (Beginning)                           |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters                             | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |

## ANSWERS:

1.

### Problem statement

Two jugs are given: one has a capacity of 4 gallons and the other has a capacity of 3 gallons. Neither jug has any measuring marks. A water pump is available to fill the jugs.

The objective is to obtain exactly **2 gallons of water in the 4-gallon jug**, using only the two jugs.

### 2. Assumptions

- The 4-gallon jug and the 3-gallon jug have no measuring marks.
- Either jug can be filled completely using the pump.
- Water can be poured from one jug into the other until the receiving jug is full or the pouring jug is empty.
- Water can be emptied onto the ground.
- No additional measuring devices are available.

### 3. State representation

Represent each state as  $(x, y)$ , where:

- $x$  = amount of water in the 4-gallon jug.
- $y$  = amount of water in the 3-gallon jug.

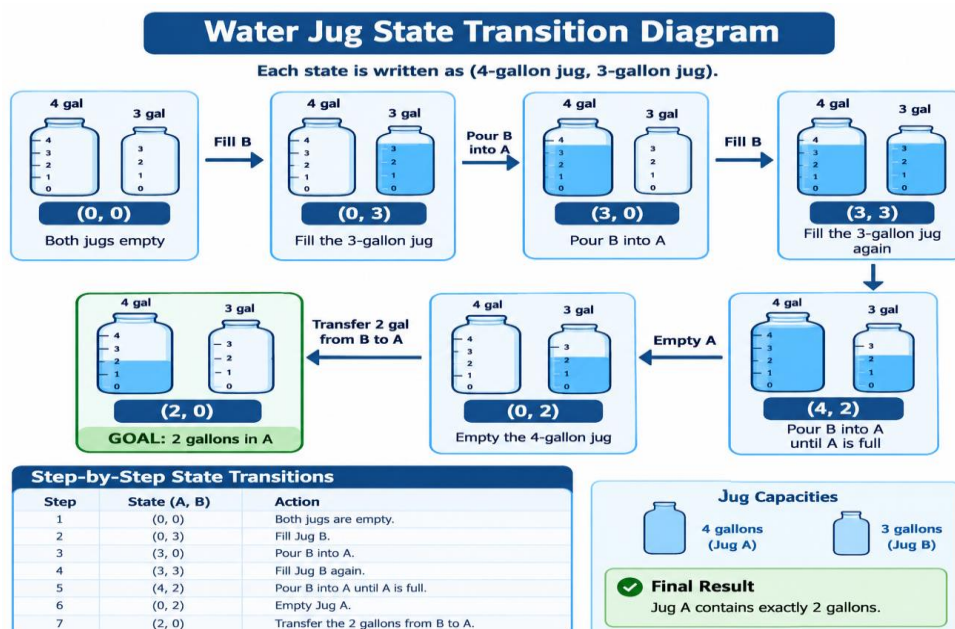
The initial state is  $(0, 0)$ , and the goal state is  $(2, 3)$ .

The goal state means that the 4-gallon jug contains exactly 2 gallons. The 3-gallon jug is also full.

The possible states satisfy:

$$0 \leq x \leq 4, 0 \leq y \leq 3$$

### 4. Diagram of the solution



Result:

The water jug problem is solved successfully. By applying the state transition operations, we reach the goal state (2, 0), where Jug A (4-gallon jug) contains exactly 2 gallons of water and Jug B (3-gallon jug) is empty.

2.

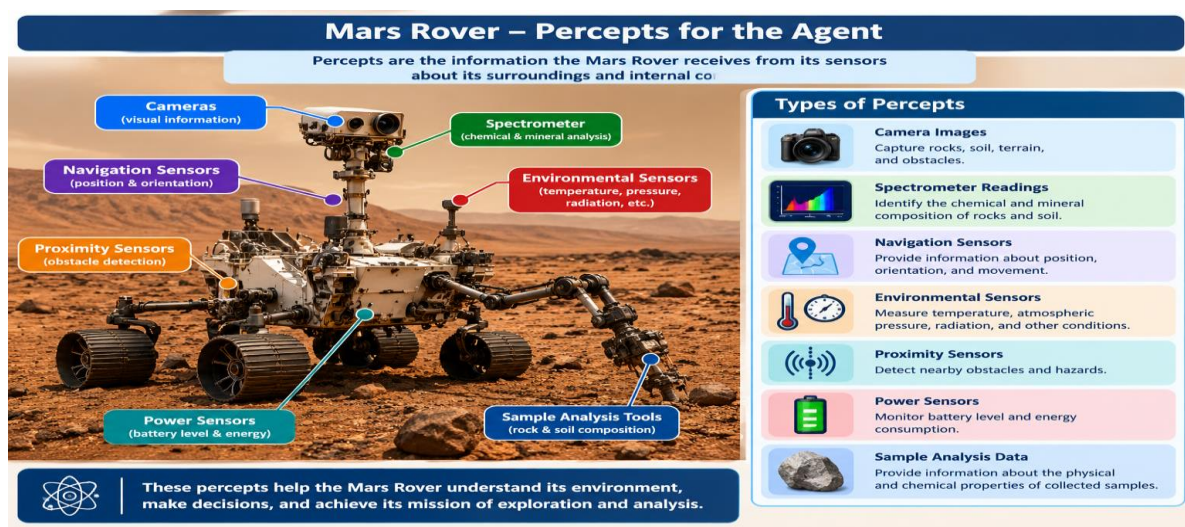
## Mars Rover – Intelligent Agent Analysis

A Mars Rover is an autonomous robotic vehicle designed to explore the surface of Mars, collect soil and rock samples, analyze their composition, and transmit scientific data to Earth. It uses sensors, artificial intelligence, and decision-making algorithms to navigate safely and perform scientific experiments.

### 1. What are the percepts for this agent?

Percepts are the information the Mars Rover receives from its sensors about its surroundings and internal condition.

- Camera images: Capture rocks, soil, terrain, and obstacles.
- Spectrometer readings: Identify the chemical and mineral composition of rocks and soil.
- Navigation sensors: Provide information about position, orientation, and movement.
- Environmental sensors: Measure temperature, atmospheric pressure, radiation, and other environmental conditions.
- Proximity sensors: Detect nearby obstacles and hazards.
- Power sensors: Monitor battery level and energy consumption.
- Sample analysis data: Provide information about the physical and chemical properties of collected samples.



## ii. What are the actions the agent can take? Characterize the operating environment.

The Mars Rover operates in the following environment:

- Partially observable: It cannot directly observe every part of the terrain or all environmental conditions.
- Partially deterministic: Some actions have predictable results, but wheel slippage, terrain changes, and equipment failures introduce uncertainty.
- Sequential: Each action affects future decisions and mission progress.
- Dynamic: Environmental conditions and hazards may change during exploration.
- Continuous: Position, speed, temperature, and other physical variables change continuously.
- Single-agent: The rover performs its own tasks, although it may coordinate with Earth-based operators or other robotic systems.
- Unknown: The rover must explore unfamiliar terrain and learn about previously unobserved conditions



## iii. What are the actions the agent can take?

The Mars Rover can perform the following actions:

1. Move and navigate: Travel forward, backward, and turn to reach target locations.

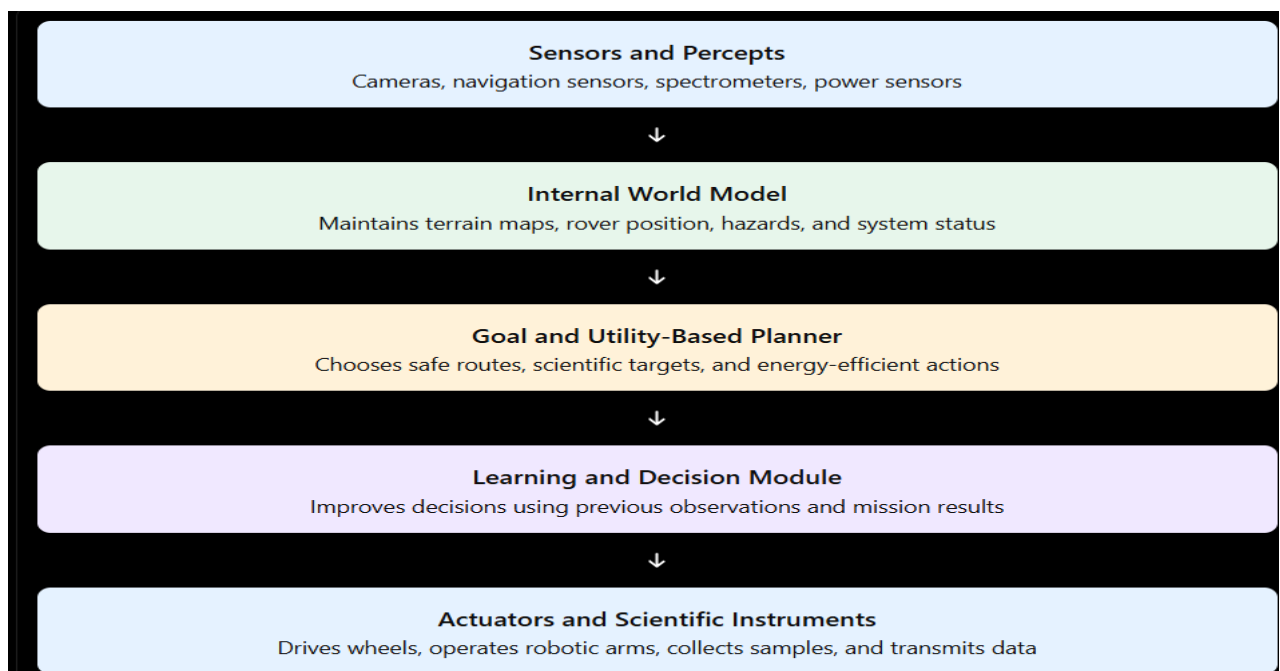
2. Avoid obstacles: Detect rocks, craters, and dangerous terrain and select a safe route.
3. Capture images: Take photographs of the Martian surface.
4. Collect samples: Use robotic arms, drills, or sampling tools to obtain soil and rock samples.
5. Analyze samples: Operate scientific instruments to determine chemical and mineral composition.
6. Transmit data: Send images, scientific results, and status reports to Earth.
7. Manage power: Adjust activities to conserve battery energy.
8. Enter safe mode: Stop risky operations when faults or unsafe conditions are detected.

#### iv. How can one evaluate the performance of the agent?

The performance of a Mars Rover can be evaluated using the following criteria:

- Scientific accuracy: Correctly identify minerals, chemicals, and sample properties.
- Mission completion: Successfully reach targets and complete assigned experiments.
- Navigation efficiency: Travel safely using minimum practical time and energy.
- Sample collection success: Collect useful samples without damaging equipment.
- Safety and reliability: Avoid collisions, prevent system failures, and operate reliably.
- Energy efficiency: Complete more tasks while consuming less power.
- Communication success: Transmit accurate scientific data to Earth.
- Autonomy: Make effective decisions with minimal human intervention.

#### vi. What sort of agent architecture is most suitable? Why?



Reasons for choosing this architecture:

- The model-based component maintains information about partially observed surroundings.
- The goal-based component plans actions to reach scientific targets.
- The utility-based component balances safety, scientific value, time, and power consumption.
- The learning component improves performance based on experience and new observations.
- The sensor and actuator feedback loop allows the rover to adapt its actions to changing conditions.

Conclusion:

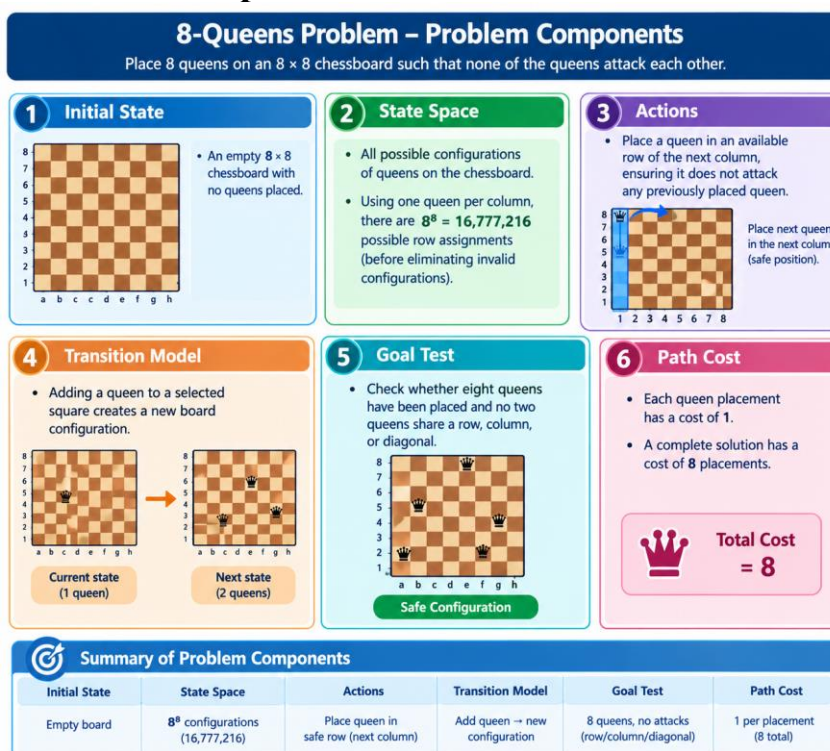
A hybrid intelligent-agent architecture is most appropriate for a Mars Rover because it combines environmental awareness, planning, optimization, and learning to explore Mars safely and efficiently with limited human assistance.

3.

## 8-Queens Problem: Problem Formulation and Search Strategies

The 8-Queens problem is a classic Artificial Intelligence search problem. The objective is to place eight queens on an  $8 \times 8$  chessboard so that no two queens attack each other. A queen can attack another queen if they share the same row, column, or diagonal.

### 1. Problem components



- **Initial state:** An empty  $8 \times 8$  chessboard with no queens placed.
- **State space:** All possible configurations of queens on the chessboard. Using one queen per column, there are  $8^8$  possible row assignments before eliminating invalid configurations.
- **Actions:** Place a queen in an available row of the next column, ensuring that it does not attack any previously placed queen.
- **Transition model:** Adding a queen to a selected square creates a new board configuration.
- **Goal test:** Check whether eight queens have been placed and no two queens share a row, column, or diagonal.
- **Path cost:** Each queen placement can have a cost of 1. A complete solution has a cost of 8 placements.

## 2. Search strategies used

### A. Breadth-First Search (BFS)

- Explores configurations level by level.
- Starts with an empty board and expands all possible placements.
- Finds a solution if one exists, but requires substantial memory.

### B. Depth-First Search (DFS)

- Places queens one at a time, exploring one configuration deeply.
- Backtracks when a queen attacks another queen.
- Requires less memory than BFS.

### C. Backtracking Search

- Places Checks whether the new position is safe.
- If no safe position exists, removes the previous queen and tries another position.
- Continues until all eight queens are placed safely.
- one queen in each column.

### D. Heuristic Search

- Uses heuristics to select promising configurations.
- For example, the Min-Conflicts heuristic moves a queen to a position with fewer conflicts.
- Can find solutions efficiently for larger variations of the problem.

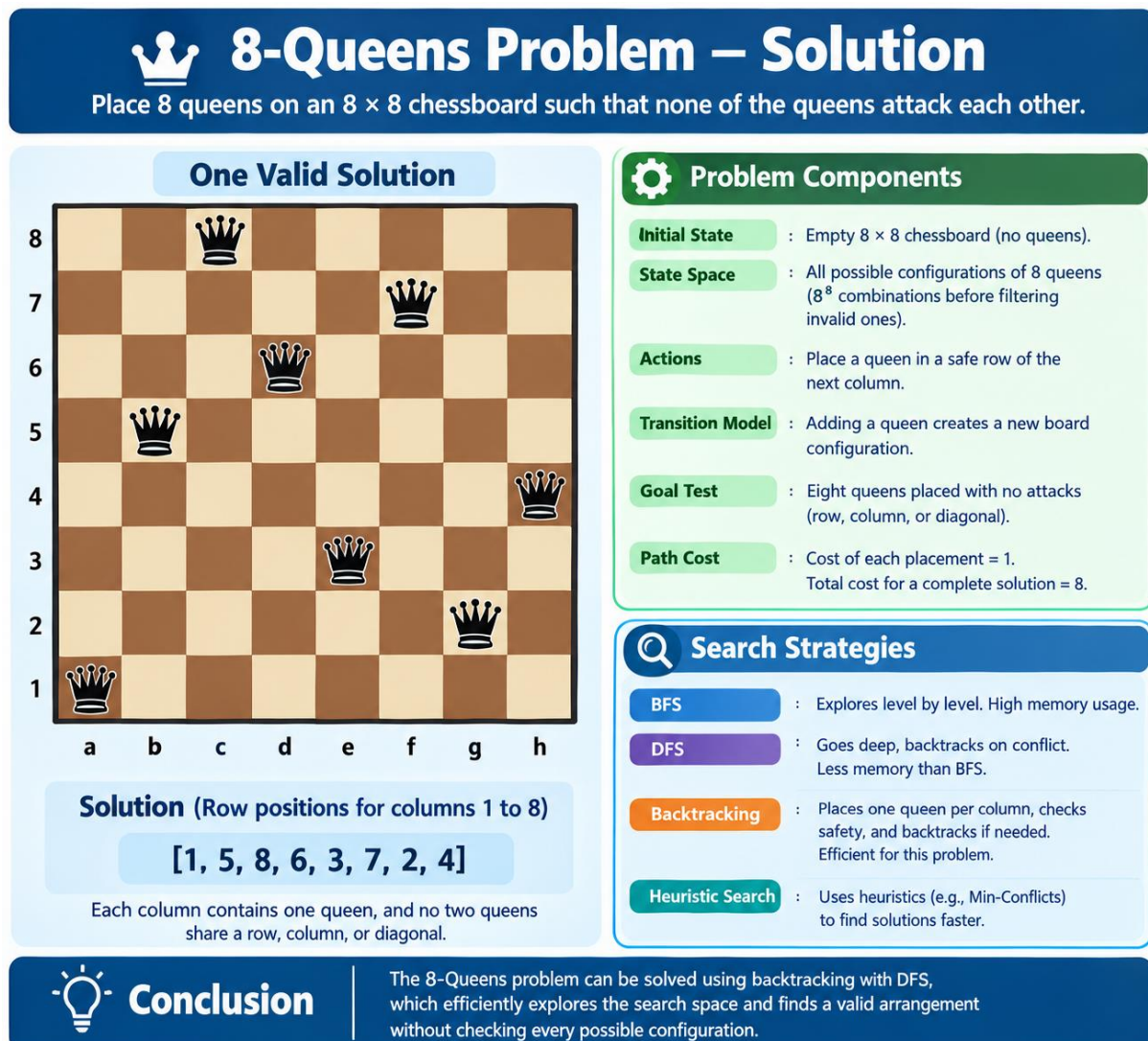
## 3. Example of a valid solution

One valid arrangement uses these row positions for columns 1 through 8:

[1, 5, 8, 6, 3, 7, 2, 4]

## 8- QUEENS SOLUTION

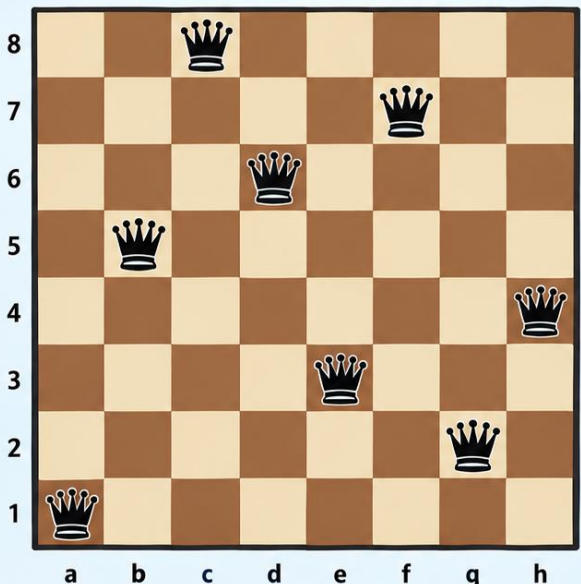
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### 8-Queens Problem – Solution

Place 8 queens on an 8 × 8 chessboard such that none of the queens attack each other.

#### One Valid Solution



**Solution (Row positions for columns 1 to 8)**  
**[1, 5, 8, 6, 3, 7, 2, 4]**

Each column contains one queen, and no two queens share a row, column, or diagonal.

#### Problem Components

- Initial State** : Empty 8 × 8 chessboard (no queens).
- State Space** : All possible configurations of 8 queens (8<sup>8</sup> combinations before filtering invalid ones).
- Actions** : Place a queen in a safe row of the next column.
- Transition Model** : Adding a queen creates a new board configuration.
- Goal Test** : Eight queens placed with no attacks (row, column, or diagonal).
- Path Cost** : Cost of each placement = 1. Total cost for a complete solution = 8.

#### Search Strategies

- BFS** : Explores level by level. High memory usage.
- DFS** : Goes deep, backtracks on conflict. Less memory than BFS.
- Backtracking** : Places one queen per column, checks safety, and backtracks if needed. Efficient for this problem.
- Heuristic Search** : Uses heuristics (e.g., Min-Conflicts) to find solutions faster.

#### Conclusion

The 8-Queens problem can be solved using backtracking with DFS, which efficiently explores the search space and finds a valid arrangement without checking every possible configuration.

## 4. Conclusion

The 8-Queens problem can be formulated as a state-space search problem. Backtracking with DFS is a particularly suitable approach because it rejects unsafe placements early, reduces unnecessary exploration, and finds a valid arrangement without checking every possible board configuration.

4.

OLA Cab Booking – Problem-Solving Agent

## 1. Identify the type of problem-solving agent

The suitable agent is a Goal-Based Problem-Solving Agent.

A goal-based agent selects actions to achieve a specific goal. In the OLA application, the customer's goal is to travel from the source location to the destination using a suitable cab.

The agent considers the pickup location, destination, cab type, estimated fare, availability, and travel time to help the customer complete the booking.

## 2. Problem formulation

- Initial state: Customer is at the pickup location.
- Goal state: Customer reaches the destination successfully.
- Actions: Enter locations, select a cab, confirm booking, and start the ride.
- State space: Possible locations, routes, cab types, and ride states.
- Transition model: Booking confirmation, cab arrival, ride commencement, and arrival at the destination.
- Path cost: Travel fare, time, distance, or a combination of these factors.

## 3. Pseudocode

BEGIN

INPUT pickup\_location

INPUT destination

IF pickup\_location = destination THEN

PRINT "Pickup and destination cannot be the same"

STOP

END IF

DISPLAY available cab types

// Mini, Micro, Sedan, Shared, Prime

INPUT selected\_cab\_type

IF selected\_cab\_type is not available THEN

PRINT "Selected cab is unavailable"

DISPLAY alternative cab types

STOP

END IF

CALCULATE estimated\_fare

CALCULATE estimated\_travel\_time

DISPLAY estimated\_fare

```

DISPLAY estimated_travel_time

INPUT customer_confirmation

IF customer_confirmation = YES THEN
    BOOK selected_cab_type
    ASSIGN driver
    TRACK cab_location

    WAIT until cab arrives
    START ride

    WHILE destination not reached
        MONITOR current_location
        FOLLOW suitable route
    END WHILE

    COMPLETE ride
    PROCESS payment
    PRINT "Destination reached successfully"
ELSE
    PRINT "Booking cancelled"
END IF

END

```

## 6. Conclusion

A goal-based problem-solving agent is suitable for the OLA cab booking application because it works toward the customer's destination goal by selecting an available cab and a suitable route. A utility-based component can also be added to compare fare, waiting time, comfort, and travel duration to recommend the best cab option.

5.

## Uniform Cost Search (UCS) – Solution

Uniform Cost Search (UCS) finds the least-cost path from the starting node  $S$  to the goal node  $G$  by expanding the node with the lowest cumulative path cost first.

### 1. Edges and costs

From the given graph:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

## 2. UCS step-by-step execution

Step Node expanded Cumulative cost Frontier after expansion (cost)

|   |   |   |                   |
|---|---|---|-------------------|
| 1 | S | 0 | A(1), G(12)       |
| 2 | A | 1 | C(2), B(4), G(12) |
| 3 | C | 2 | D(3), G(4), B(4)  |
| 4 | D | 3 | G(4), B(4)        |
| 5 | G | 4 | Goal reached      |

Note: Nodes with equal costs can be expanded in either order. The solution cost remains the same.

## 3. Path cost calculations

Possible routes include:

- $S \rightarrow G = 12$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = 1 + 3 + 3 + 3 = 10$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 1 + 1 + 1 + 3 = 6$
- $S \rightarrow A \rightarrow C \rightarrow G = 1 + 1 + 2 = 4$

## 4. Final answer

Optimal path found

$S \rightarrow A \rightarrow C \rightarrow G$

Total cost

**4**

Conclusion:

Uniform Cost Search finds the least-cost path  $S \rightarrow A \rightarrow C \rightarrow G$ , with a total travel cost of 4.